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Liu et al.

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(54) **SNAP ON CURRENT SENSOR DESIGN**

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B60R 16/023 (2006.01)

(52) **U.S. Cl.**
CPC ***B60R 16/0239*** (2013.01)

(58) **Field of Classification Search**
CPC B60R 16/0239
USPC 361/752, 728, 796, 800
See application file for complete search history.

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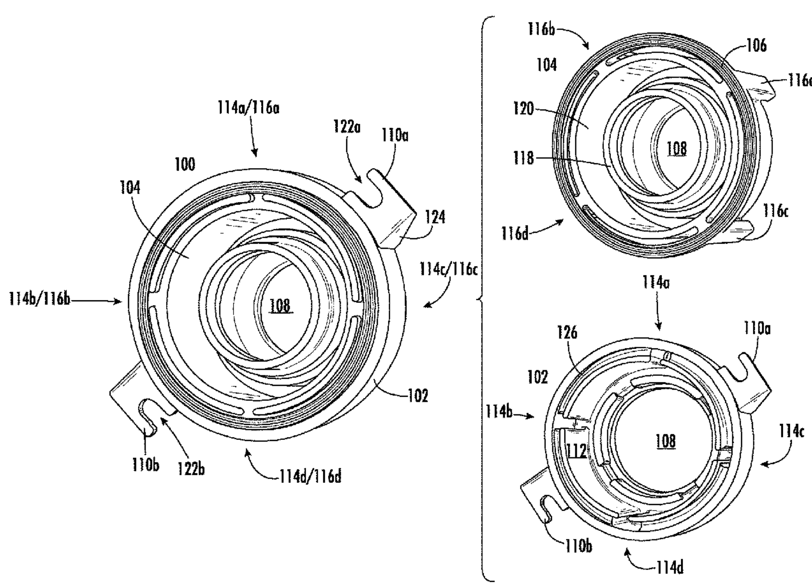
(57) **ABSTRACT**

A seal for securing a cable to a fuse box includes a plastic body and a rubber insert. The plastic body has a receiving cavity along an inner surface. The rubber insert features an outside structure to join with the inner surface of the plastic body, an inside structure to surround the cable, and a fitting rib along an outside surface of the outside structure, the fitting rib to occupy the receiving cavity. The inside structure of the rubber insert moves in response to movement of the cable while the outside structure does not move.

5 Claims, 10 Drawing Sheets



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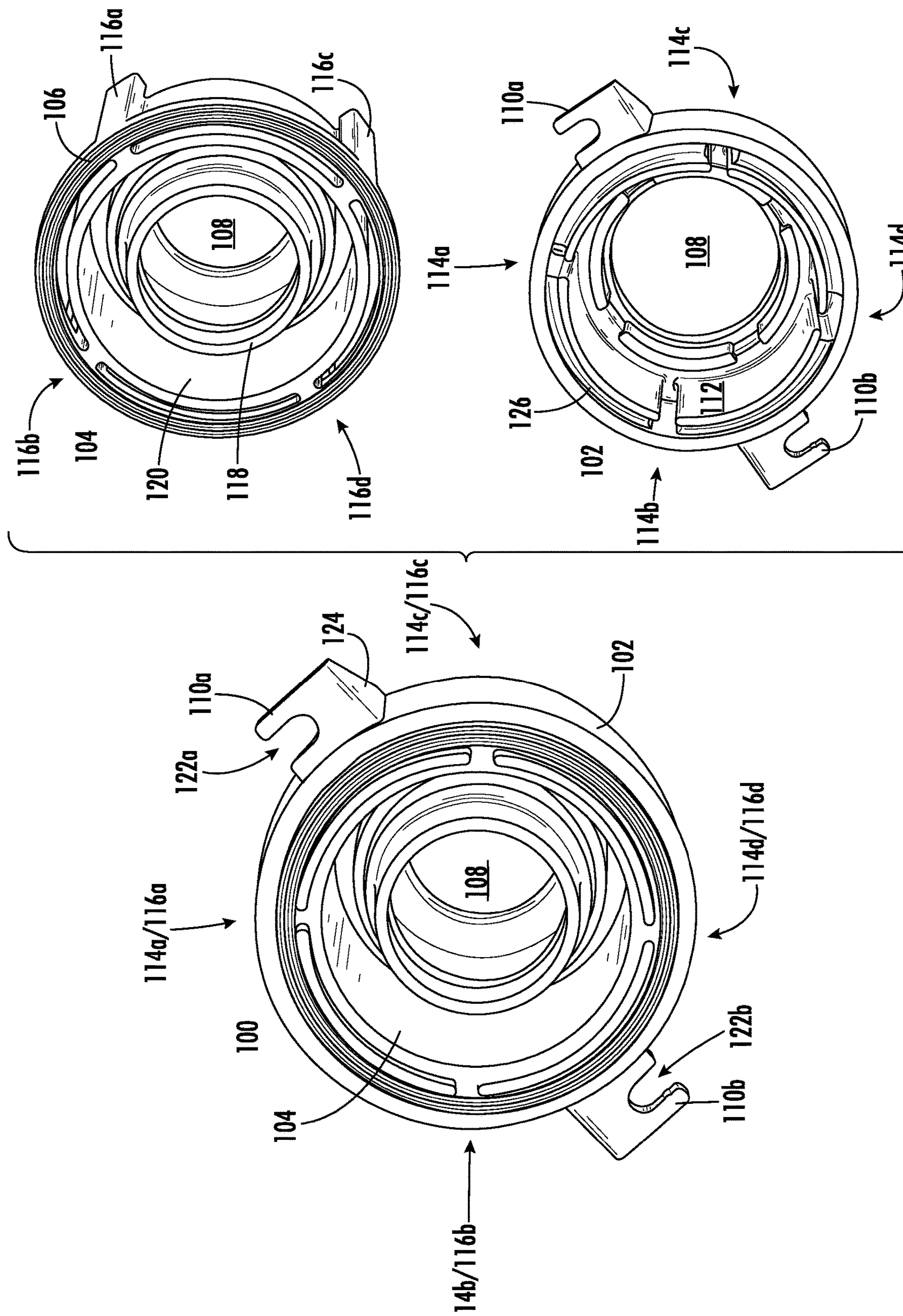
116a
116c
106
108
116b
104
120
118
116d
114c
110a
114a
114d
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114b
110b
114c/116c
124
122a
114a/116a
114d/116d
100
122b
114b/116b

FIG. 1

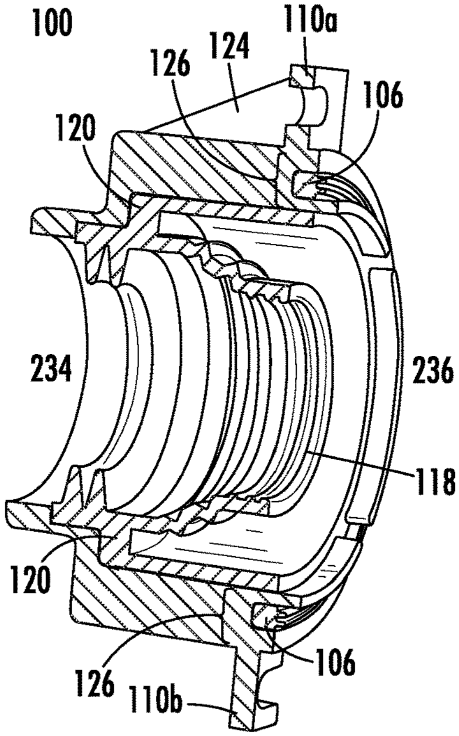

FIG. 2A

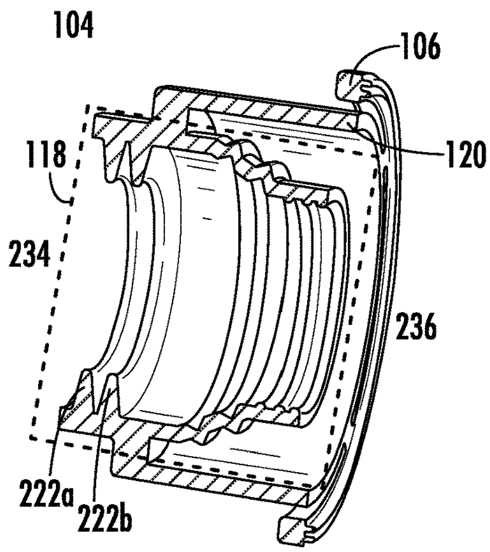

FIG. 2B

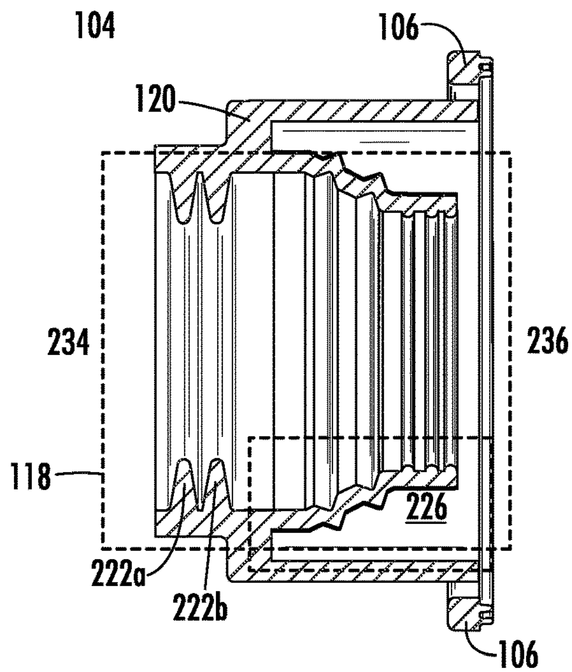

FIG. 2C

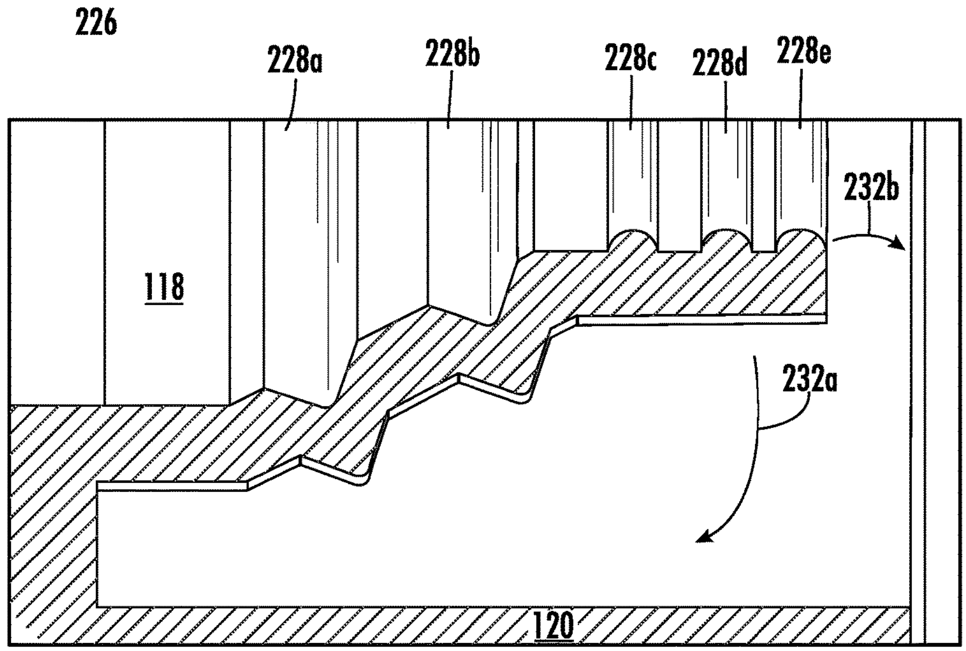
226
228a
228b
228c
228d
228e
232b
118
232a
120

FIG. 2D

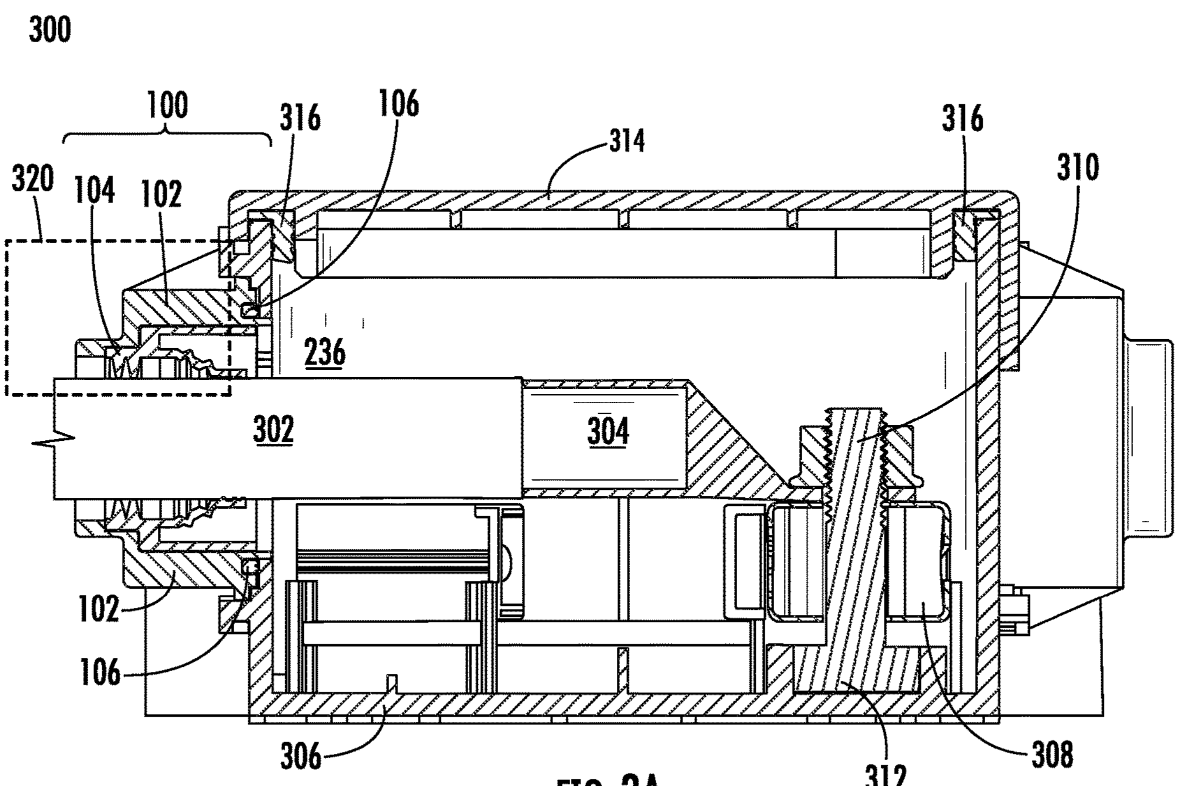
300
100
316
106
314
316
310
320
104
102
236
302
304
102
106
306
312
308

FIG. 3A

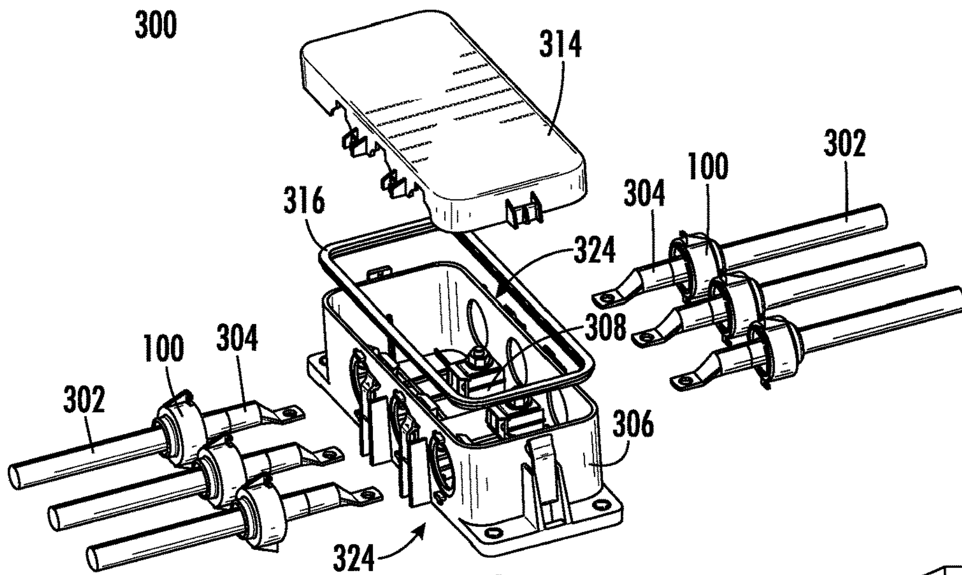

FIG. 3B

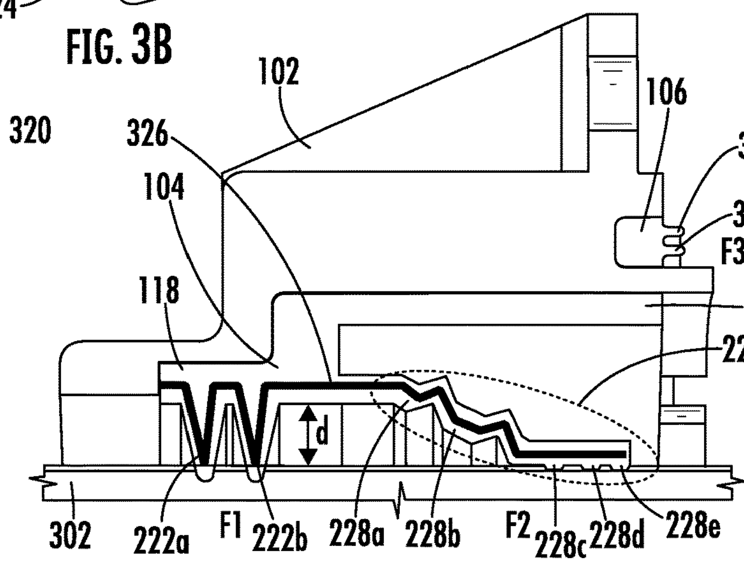

FIG. 3C

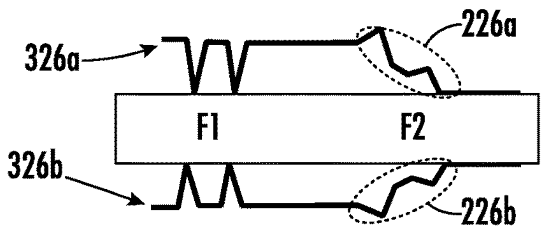

FIG. 3D

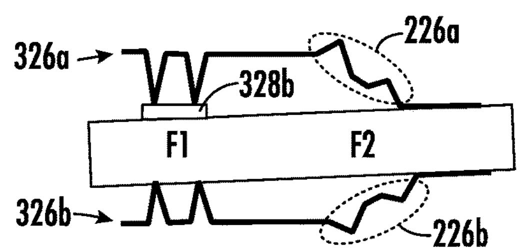

FIG. 3F

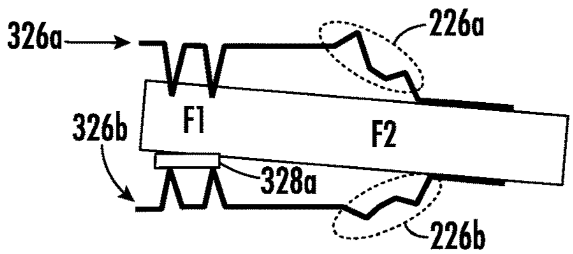

FIG. 3E

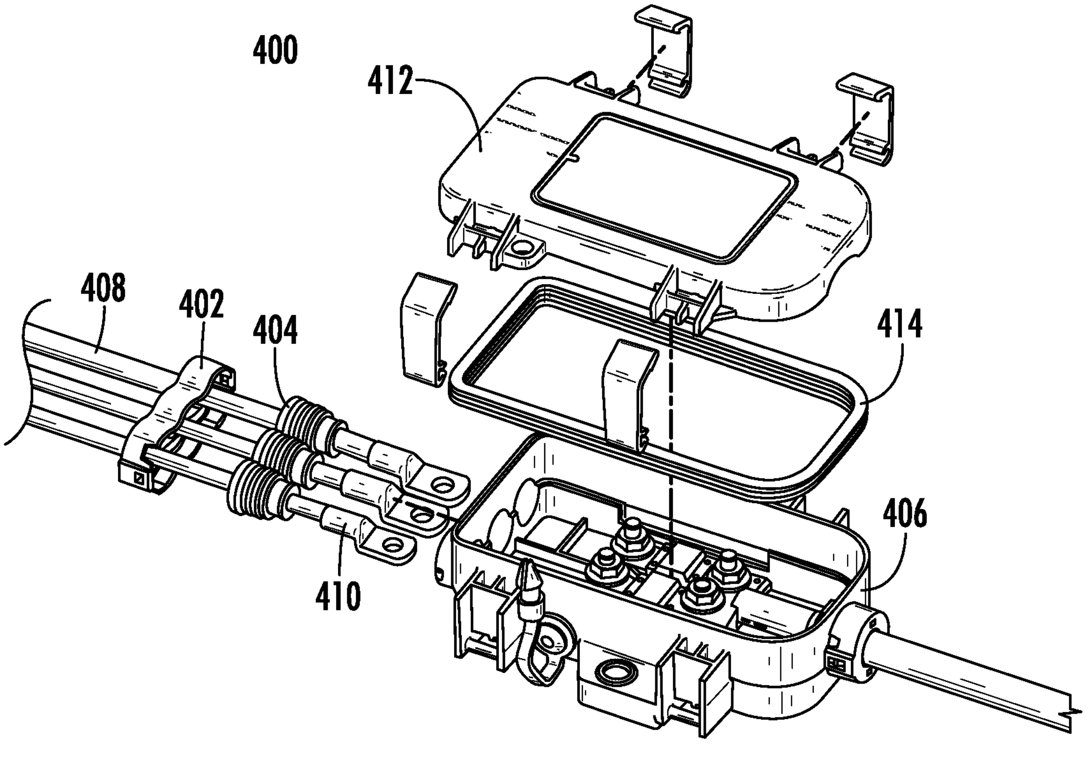

FIG. 4A
(PRIOR ART)

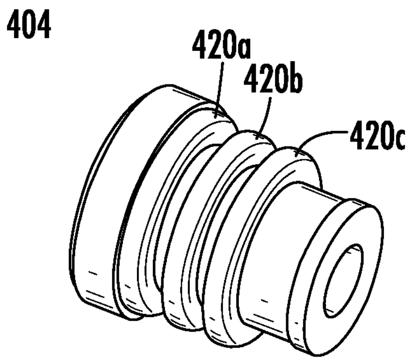

FIG. 4B
(PRIOR ART)

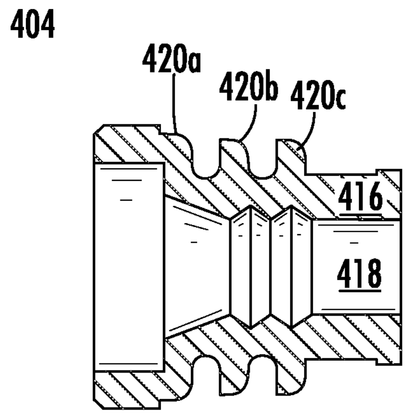

FIG. 4C
(PRIOR ART)

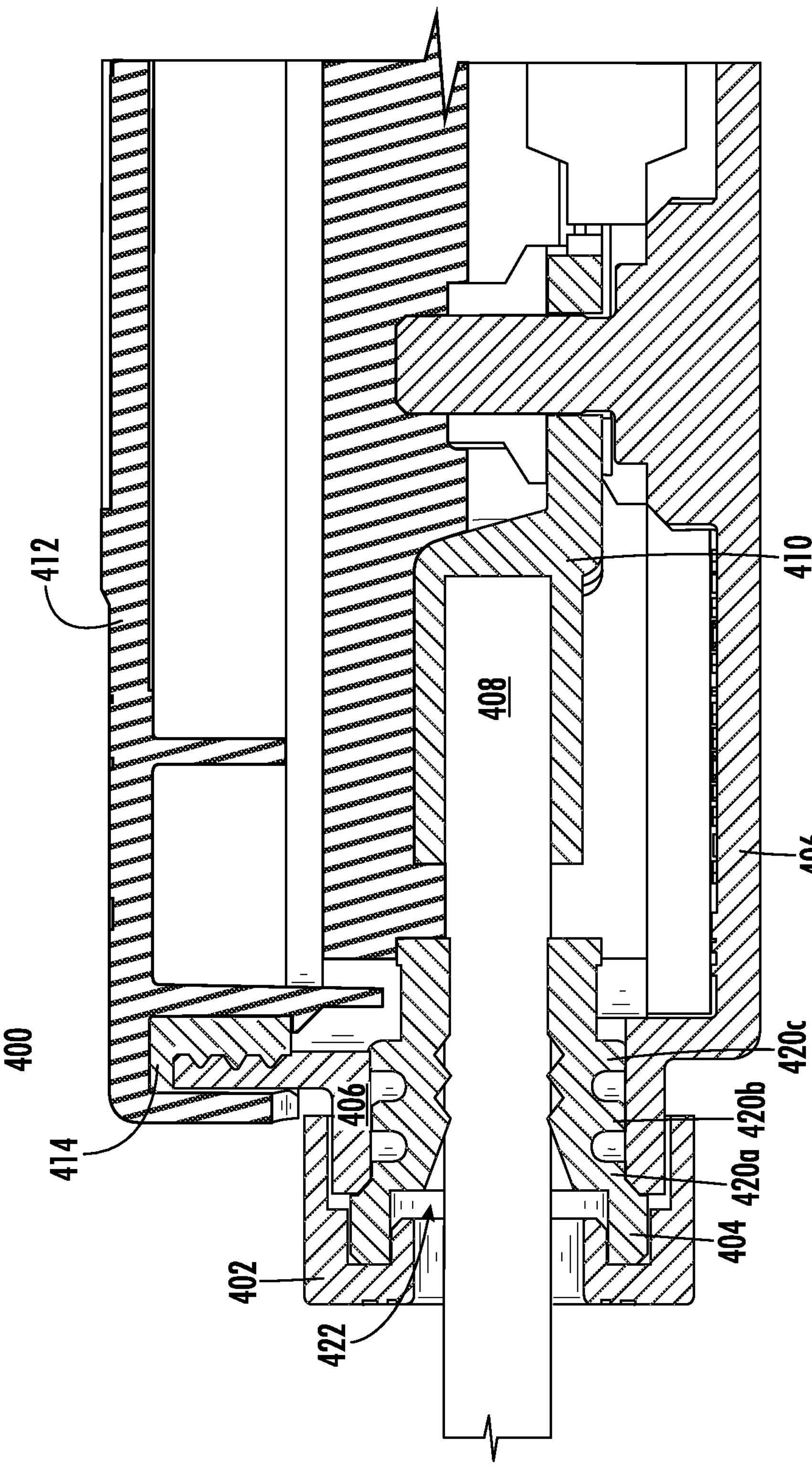

FIG. 4D
(PRIOR ART)

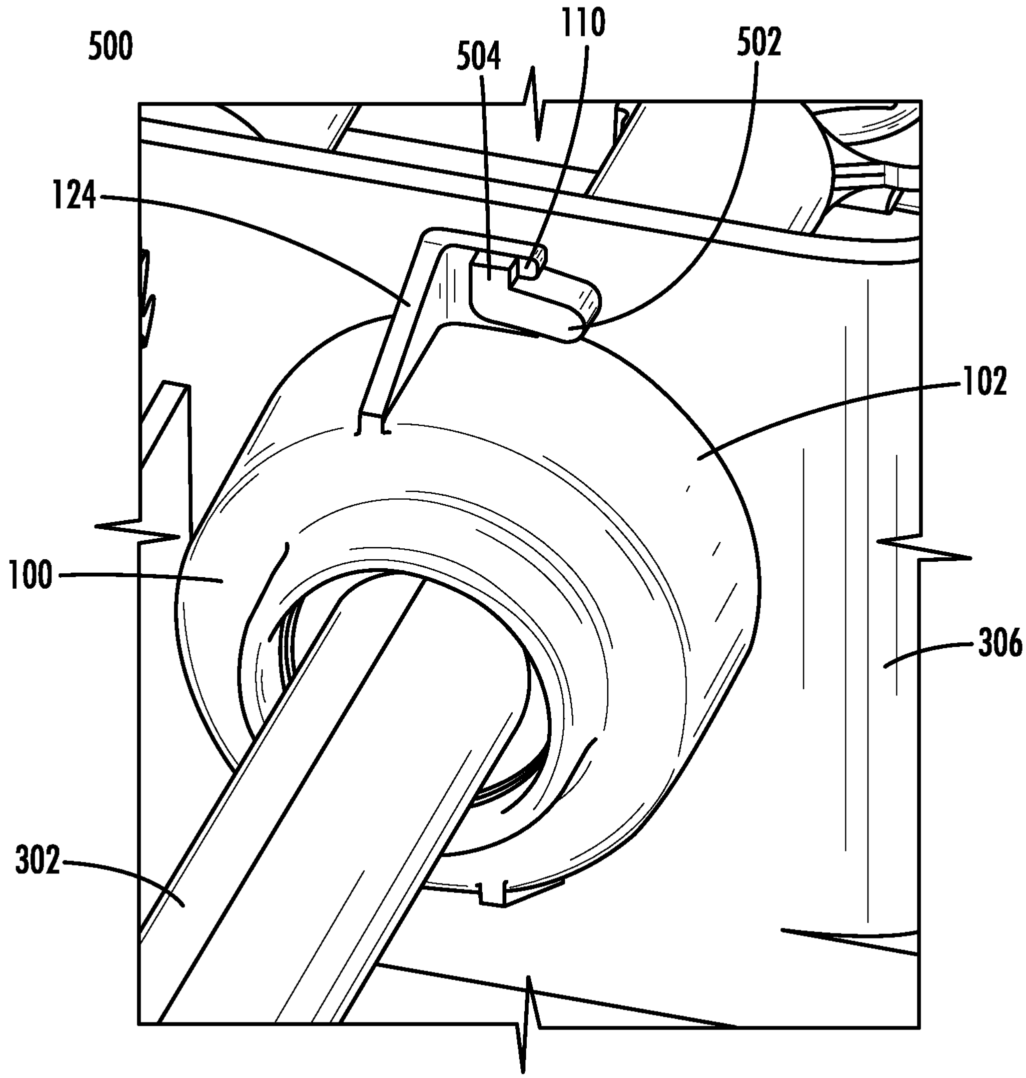

FIG. 5

600

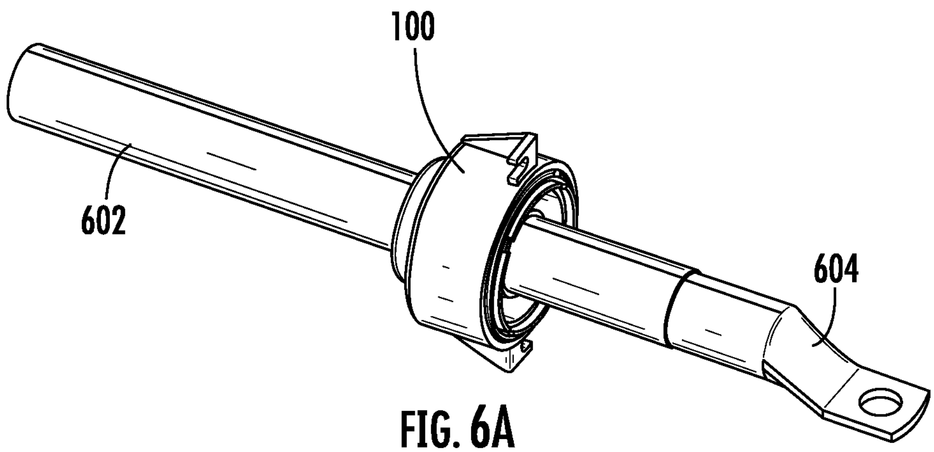

FIG. 6A

610

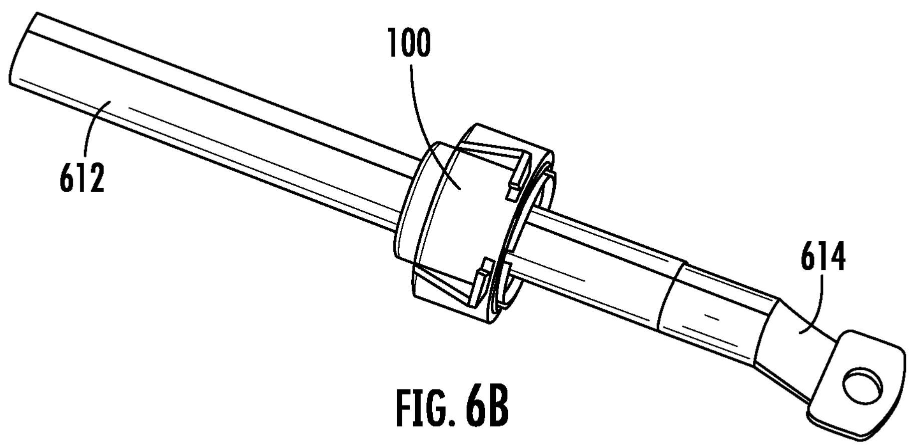

FIG. 6B

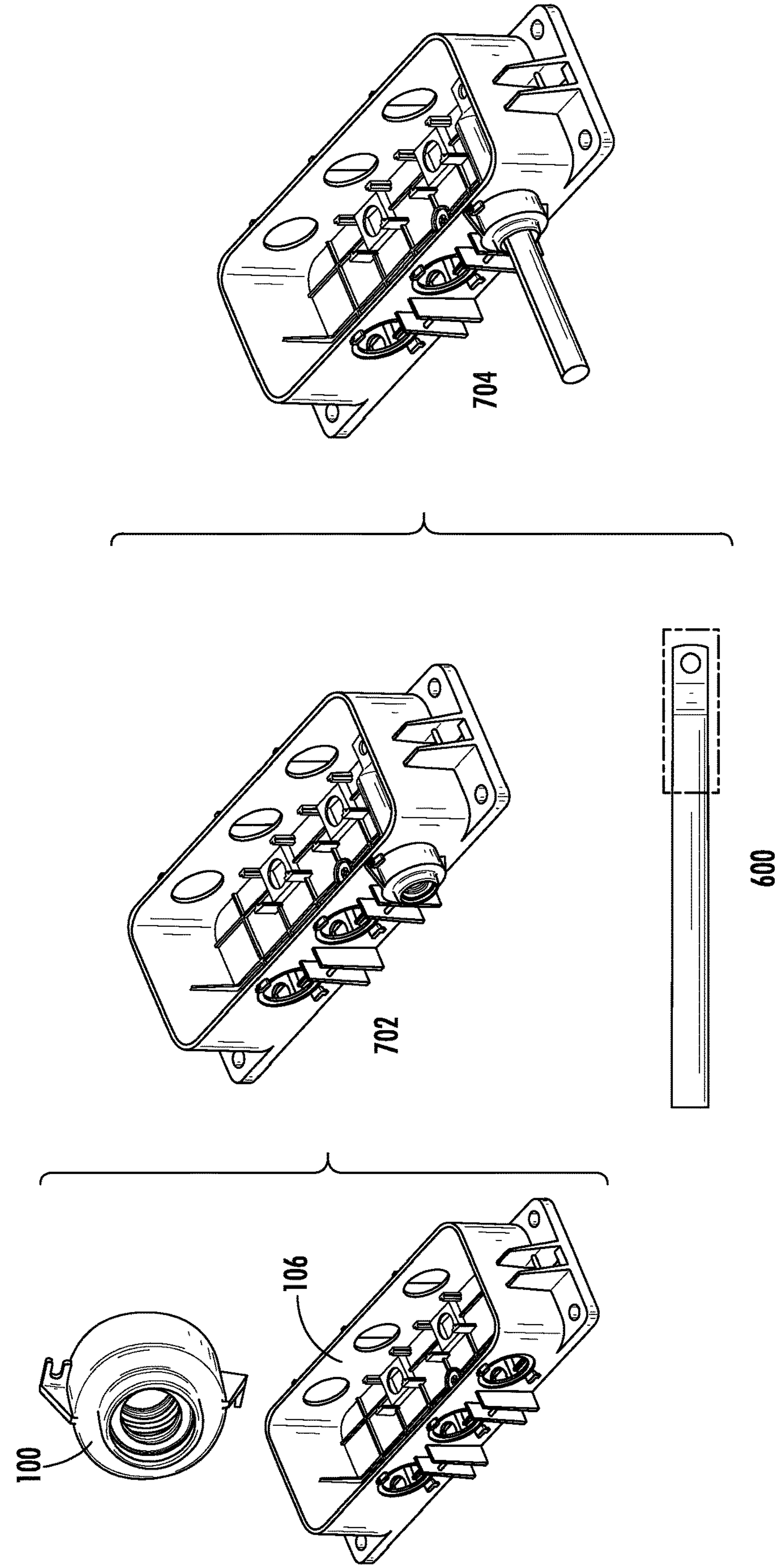

FIG. 7

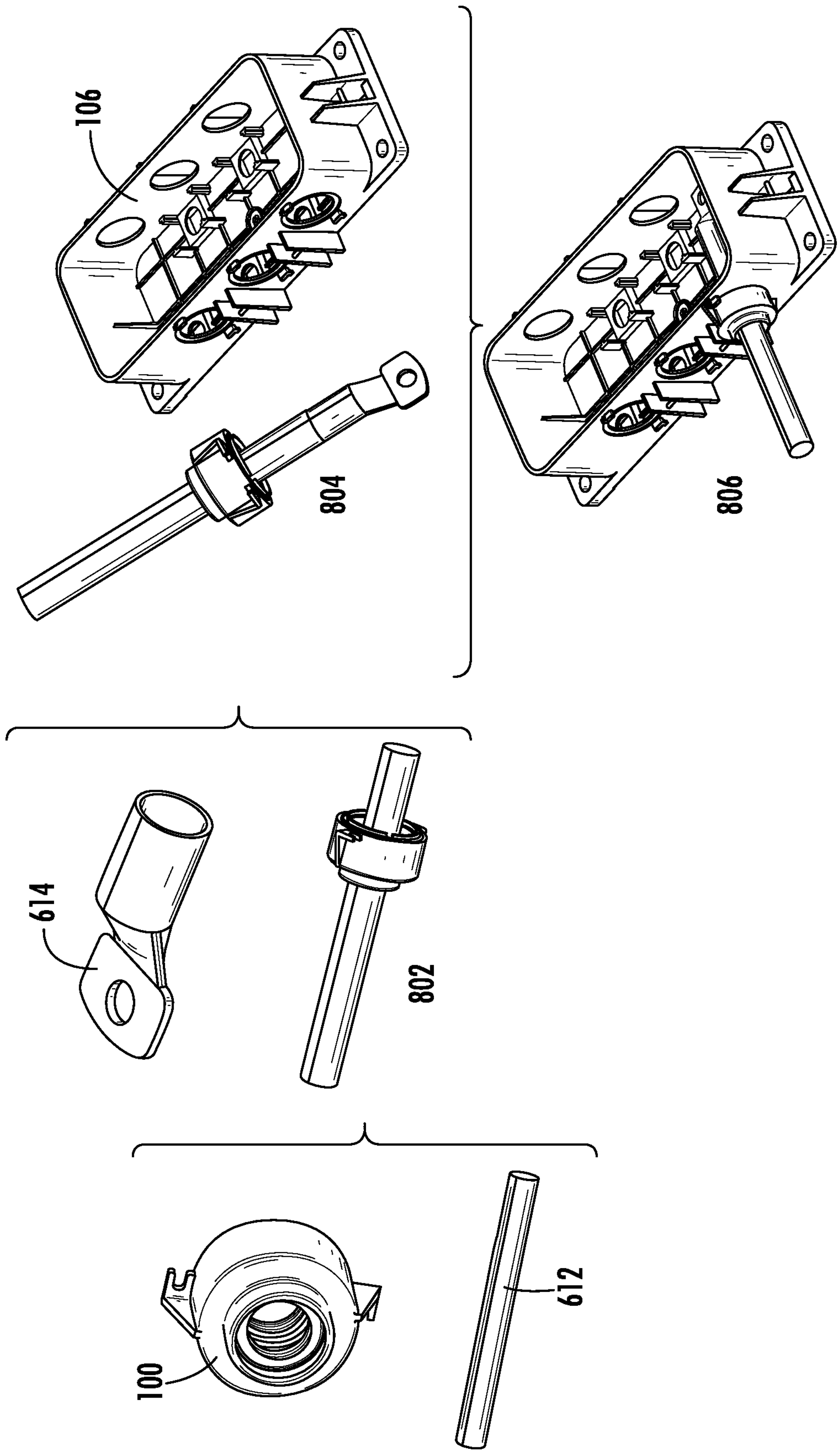

FIG. 8

SNAP ON CURRENT SENSOR DESIGN

FIELD OF THE DISCLOSURE

Embodiments of the present disclosure relate to fuse boxes and, more particularly, to fuse boxes that satisfy IP67/IP69 Ingress Protection ratings.

BACKGROUND

Vehicles are equipped with a variety of electrically powered equipment which may be protected with one or more fuses. The fuses may be collected in groups and housed in fuse boxes.

The fuses are connected to other circuits within the vehicles by cables. Thus, the cables are inserted into the fuse boxes and connected to the fuses. Manufacturing a fuse box that has both IP67 (total protection from dust and protected from temporary liquid immersion) and IP69K (proven to resist ingress of high temperature and pressure wash) Ingress Protection ratings is challenging because the fuses are accessed during maintenance, and thus cannot be permanently sealed within the fuse box.

It is with respect to these and other considerations that the present improvements may be useful.

SUMMARY

This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended as an aid in determining the scope of the claimed subject matter.

An exemplary embodiment of a seal in accordance with the present disclosure may include a plastic body and a rubber insert. The plastic body has a receiving cavity along an inner surface. The rubber insert features an outside structure to join with the inner surface of the plastic body, an inside structure to surround the cable, and a fitting rib along an outside surface of the outside structure, the fitting rib to occupy the receiving cavity. The inside structure of the rubber insert moves in response to movement of the cable while the outside structure does not move.

An exemplary embodiment of a fuse box assembly in accordance with the present disclosure may include a housing, a fuse, a cover, and a seal. The housing has an opening for receiving a cable. The fuse is connected to a terminal of the cable. The cover fits over the housing and seals the fuse inside the housing. The seal fits into the opening and holds the cable against the housing and prevents ingress of water or dust into the housing. The seal consists of an overmolded combination of a plastic body and a rubber insert. The plastic body features a lock mechanism to join the seal to the housing. The rubber insert features an outside structure and an inside structure, with the outside structure being connected to the plastic body and being stationary and the inside structure to surround the cable and being movable within the outside structure. The inside structure simultaneously elongates and compresses in response to movement of the cable.

An exemplary embodiment of rubber insert to be overmolded with a plastic body to form a water-tight seal in accordance with the present disclosure may include an outside structure and an inside structure. The outside structure is to be joined with an inner surface of the plastic body and is stationary. The inside structure is to circumferentially grip around a cylindrical cable. The inside structure includes a pair of ribs disposed circumferentially on an insertion side of the inside structure. The pair of ribs compress in response to insertion of the cylindrical cable. The inside structure further includes multiple grippers disposed circumferentially on a terminal side of the inside structure. The multiple grippers are movable in a space between the inside structure and the outside structure.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a diagram illustrating a seal for use with a fuse box, in accordance with exemplary embodiments;

FIG. **2**A-**2**D are diagrams illustrating the seal of FIG. **1**, in accordance with exemplary embodiments;

FIGS. **3**A-F are diagrams illustrating a fuse box assembly using the seal of FIG. **1**, in accordance with exemplary embodiments;

FIGS. **4**A-**4**D are diagrams illustrating a fuse box assembly, in accordance with the prior art;

FIG. **5** is a diagram illustrating a locked seal assembly used by the seal of FIG. **1**, in accordance with exemplary embodiments;

FIGS. **6**A-**6**B are diagrams illustrating two cables secured using the seal of FIG. **1**, in accordance with exemplary embodiments;

FIG. **7** is a diagram illustrating process steps for attaching a cable with a small terminal to a fuse box using the seal of FIG. **1**, in accordance with exemplary embodiments; and

FIG. **8** is a diagram illustrating process steps for attaching a cable with a large terminal to a fuse box using the seal of FIG. **1**, in accordance with exemplary embodiments.

DETAILED DESCRIPTION

A novel seal is disclosed herein for providing IP69 at the point at which a cable is inserted into a fuse box assembly. The seal is an overmolded part consisting of a rigid plastic body and a flexible rubber insert. The rubber insert includes several features that add surface area to its inside structure and further increase its flexibility. A portion of the rubber insert can simultaneously elongate and compress in response to movement of the cable, thus ensuring IP69 protection is maintained, even from high-pressure and/or high-temperature sprays of water.

For the sake of convenience and clarity, terms such as "top", "bottom", "upper", "lower", "vertical", "horizontal", "lateral", "transverse", "radial", "inner", "outer", "left", and "right" may be used herein to describe the relative placement and orientation of the features and components of the electrical box, each with respect to the geometry and orientation of other features and components of the electrical box appearing in the perspective, exploded perspective, and cross-sectional views provided herein. Said terminology is not intended to be limiting and includes the words specifically mentioned, derivatives therein, and words of similar import.

FIG. **1** is a representative drawing of a seal **100** for use with a fuse box, according to exemplary embodiments. The seal **100** is an overmolded component made of a plastic material and a rubber material. Overmolding is a technique that allows the combination of multiple materials into a unitary structure. The seal **100** is thus an overmolded combination of a plastic body **102** and rubber insert **104**. In exemplary embodiments, the plastic body **102** is a thermoplastic, such as PA66, and the rubber insert **104** is silicon rubber. Both the plastic body **102** and the rubber insert **104** include an opening **108** through which a cable (not shown) is disposed, for connecting and securing the cable to a fuse box, such as the fuse box assembly **300** in FIGS. **3**A-B, below.

In exemplary embodiments, the plastic body **102** includes an inner surface **112** for receiving the rubber insert **104** while the rubber insert includes an inside structure **118** and an outside structure **120**. The inner surface **112**, the inside structure **118**, and the outside structure **120** are substantially cylindrical in shape. In exemplary embodiments, the inner surface **112** is further shaped such that the outside structure **120** of the rubber insert **104** fits snugly into the inner surface of the plastic body **102**.

The plastic body **102** features two lock mechanisms **110***a* and **110***b* (collectively, "lock mechanism(s) **110**") for securing the seal **100** against the housing of a fuse box. Lock mechanism **110***a* includes an opening **122***a* and lock mechanism **110***b* includes an opening **122***b* (collectively, "openings **122**"). Although two lock mechanisms are shown, the plastic body **102** includes a single lock mechanism, in one embodiment. Only visible for the lock mechanism **110***a*, a corner piece is disposed orthogonal to the opening **122**, which stabilizes the opening, in some embodiments. The openings **122** are designed to clasp with respective engagement mechanisms of a fuse box assembly. The lock mechanism **110** is described in more detail in conjunction with FIG. **5**, below.

The plastic body **102** also includes receiving cavities **114***a-d* (collectively, "receiving cavities **114**") disposed along the inner surface **112** while the rubber insert **104** includes fitting ribs **116***a-d* disposed on an outside surface of the outside structure **120** (collectively, "fitting ribs **116**"). Although the plastic body **102** features four receiving cavities **114** and the rubber insert **104** features four fitting ribs **116**, the seal **100** may include more or fewer receiving cavities and respective fitting ribs. Because the seal **100** is an overmolded part, the fitting ribs **116** of the rubber insert **104** occupy respective receiving cavities **114** of the plastic body **102**. The rubber insert **104** further includes a ring **106** for fitting the rubber insert with the plastic body **102**. In exemplary embodiments, the ring **106** is connected to each of the fitting ribs **116**. The ring **106** is discussed in more detail in FIGS. **2**A-**2**D.

The International Electrotechnical Commission promulgates Ingress Protection (IP) ratings to classify the sealing effectiveness of electrical enclosures from foreign matter such as water and dust. The IP rating is a two-digit number, where the first digit pertains to ingress of solids and the second digit pertains to the ingress of water. Having an IP rating of "6" in the first digit indicates that the enclosure is "dust tight", that is, no dust will enter the enclosure. An IP rating of "9" in the second digit indicates protection from close-range water pressure as well as high temperature water. In exemplary embodiments, the seal **100** used to connect or couple a cable to an electrical box, provides IP69 protection of the box from ingress of both dust and water, even high-pressure and high-temperature water. The seal **100** may therefore be useful in applications in the automotive industry.

FIGS. **2**A-**2**D are representative drawings of the seal **100** of FIG. **1**, along with further detail about the rubber insert **104**, according to exemplary embodiments. FIG. **2**A is a perspective cutaway view of the seal **100**; FIG. **2**B is a perspective cutaway view of the rubber insert **104**; FIG. **2**C is a side cutaway view of the rubber insert **104**; and FIG. **2**D is a detailed view of a portion of the inside structure **118**. In describing the components of the overmolded seal **100**, particularly the plastic body **102** and the rubber insert **104** components, reference is made to an insertion side **234** and a terminal side **236**. The insertion side **234** is the side of the seal **100** from which the cable will be inserted through the seal. The terminal side **236** is the side of the seal **100** closer to the terminal of the cable (see, e.g., FIG. **3**A). Both the insertion side **234** and the terminal side **236** are indicated in FIGS. **2**A-**2**D.

In FIG. **2**A, the plastic body **102** of the seal **100** is shown along with the locking mechanisms **110**, with another view of the corner piece **124** being visible for the lock mechanism **110***a*. In exemplary embodiments, the lock mechanism **110** is located on the terminal side **236** of the plastic body **102**. In exemplary embodiments, the corner piece **124** is perpendicular to the circumferential edge of the plastic body **102** while the lock mechanism **110** is orthogonal to the corner piece. Inside the plastic body **102**, the ring **106**, the inside structure **118**, and the outside structure **120** of the rubber insert **104** are visible, where the ring **106** is on the terminal side **236** of the rubber insert. In exemplary embodiments, also on the terminal side **236**, the plastic body **102** further includes a receiving groove **126** shaped so that the ring **106** of the rubber insert **104** fits therein.

In exemplary embodiments, the rubber insert **104** is fixed with wire (not shown) to ensure a good seal with the cable. Nevertheless, the rubber insert **104** is elastic and flexible enough to move in response to movement of the cable once inserted in the seal **100**. In FIGS. **2**B and **2**C, the inside structure **118** of the rubber insert **104** is outlined with a dashed rectangle. The outside structure **120** is connected to the ring **106**. Also visible in FIG. **2**A, there exists some space between the inside structure **118** and the outside structure **120** of the rubber insert **104**. In exemplary embodiments, this space allows the inside structure **118** to move somewhat within the outside structure **120** while the outside structure stays stationary within the plastic body **102**. The flexibility of the inside structure **118** thus enables movement of the cable once inserted through the seal **100** while the rigidity of the outside structure helps to maintain the integrity of the seal within the opening of a fuse box. In contrast to prior art seal mechanisms, the movement of the cable with the seal **100** will not allow ingress of materials such as dust and water into the fuse box to which the cable is connected.

In exemplary embodiments, the inside structure **118** of the rubber insert **104** includes ribs **222***a* and **222***b* (collectively, "ribs **222**"), located on the insertion side **234**, as well as a gripper section **226** of additional surfaces, located on the terminal side **236**. The gripper section **226** is outlined at the bottom of FIG. **2**C, and, because the gripper section **226** and the ribs **222** are circumferentially disposed around the inside structure **118** of the rubber insert **104**, both features are also shown at the top of the inside structure. The gripper section **226** is shown in more detail in FIG. **2**D. Both the ribs **222** and the gripper section **226** increase the surface area and add flexibility to the inside structure **118** of the rubber insert **104**. In exemplary embodiments, the gripper section **226** of the inside structure **118** of the rubber insert **104** is designed to either elongate or compress in response to movement of the cable being connected by the seal **100**.

The gripper section **226** features grippers **228***a-e* formed along the inner surface of the inside structure **118** (collectively, "grippers **228**"). Both the ribs **222** and the grippers **228** are circumferentially disposed around the entire inside surface of the inside structure **118** and thus fit tightly and circumferentially against the cable inserted into the seal **100**. In some embodiments, as illustrated in FIG. **3**B, below, the ribs **222** and grippers **228** overlap with the cable, with the ribs **222** connecting with the cable at the insertion side **234** and the grippers connecting with the cable at the terminal side **236**. The inside structure **118** of the rubber insert **104** thus creates a tight seal against the cable, in some embodiments, ensuring that no dust or water can get into the fuse box.

In exemplary embodiments, the gripper section **226** of the inside structure **118** is elastic and thus able to move within the outside structure **120**, while the outside structure remains stationary. This is useful if the cable being held by the seal **100** moves. Spaces **232***a* and **232***b* are shown, which are spaces between the inside structure **118** and the outside structure **120** (collectively, "spaces **232**"). The spaces **232** enable the inside structure **118** to elongate or compress in response to movement of the cable inside the seal **100**. The space **232***a* particularly shows that there is considerable movement available within the inside structure **118** such that contact with the outside structure **120** of the rubber insert **104** is avoided, in some embodiments. This ensures that the outside structure **120** will remain stationary while the inside structure **118** is able to compress and elongate in response to movement of the cable, in exemplary embodiments.

FIGS. **3**A-**3**F are representative drawings of a fuse box assembly including the seal **100** of FIG. **1**, according to exemplary embodiments. FIG. **3**A is a side cutaway view of the fuse box assembly **300** with the seal **100**; FIG. **3**B is a perspective exploded view of the fuse box assembly **300**; FIG. **3**C is a side cutaway view of an inset of the fuse box of FIG. **3**A; and FIGS. **3**D-**3**F are side views showing response of the seal to cable movement.

The fuse box assembly **300** features a cable **302** and terminal **304** to be connected to a fuse **308** using a screw bolt **312** and nut **310**. This arrangement enables an electrical connection to be established between the fuse **308** and the circuit to which the cable **302** is attached. The fuse box assembly **300** further includes a housing **306** and a cover **314**, which is connected to the housing by a gasket **316**. The exploded view of FIG. **3**B shows that the gasket, made from a rubber or other elastomeric material, is arranged circumferentially between the housing **306** and the cover **314**, both of which are made of a harder, plastic material. The housing **306** includes openings **324** through which the cables **302** are inserted to connect to respective fuses **308**.

As illustrated in the side cutaway view of FIG. **3**A, the seal **100**, including the plastic body **102** and the rubber insert **104**, surrounds the cable **302** to hold the cable in place against the housing **306** in the fuse box assembly **300**. Although the terminal **304** of the cable **302** is connected to the fuse **308** using the screw bolt **312** and nut **310**, the seal **100** provides additional insurance that the cable **302** will remain in place in the fuse box assembly **300**. Further, in exemplary embodiments, the seal **100** also prevents ingress of water or dust into the housing **306**, thus protecting the fuse **308**.

An inset **320** of the seal **100** is denoted with a dashed rectangle and is further illustrated in FIG. **3**C. A gradient **326** of the inside structure **118** of the rubber insert **104** is shown, with the ribs **222** being disposed the insertion side **234** and the grippers **228** on the terminal side **236**. The ribs **222***a* and **222***b*, as well as grippers **228***c*, **228***d*, and **228***e* overlap somewhat with the cable **302**, indicating that the inside structure **118** of the rubber insert **104** tightly grips the cable when the cable is inserted, in some embodiments. Further, in some embodiments, the rubber insert **104** of the seal **100** is behind the opening of the housing **306**, preventing even high-pressure water from getting into the fuse box assembly **300**.

Considering that the fuse box assembly **300** may be a component under the hood of a vehicle, as one example, the cable **302** may become bent or off-center in the assembly, causing the gripper section **226** of the inside structure **118** to move. Events such as movement following assembly and normal vehicle use (e.g., driving) may cause the cable **302** to move. Where the cable **302** becomes bent or in an off-center position, the grippers **228***a* and **228***b* may also touch the cable. The inset **320** thus shows that the inside structure **118** features several structures that, given any position of the cable **302**, will alternatively elongate or compress to maintain contact with the cable, thus protecting the inside of the fuse box assembly **300** from ingress of dust, water, grease, or other foreign objects. Put another way, the design of the rubber insert **104** of the seal **100** ensures that gaps between the seal and the cable **302** do not occur, in some embodiments.

In FIGS. **3**C-**3**F, protective positions are denoted by F1, F2, and F3, respectively. In exemplary embodiments, the inside structure **118** of the rubber insert **104** is designed like a spring that can both elongate and compress, in response to movement of the cable **302**. Being cylindrical in shape, the cable **302** can move in all directions, as there are up to 360° of movement from the original insertion position, not simply upward or downward positions. Similarly, the seal **100** is cylindrical in shape and able to respond to movement of the cable **302** in all directions. Nevertheless, FIGS. **3**D-**3**F are provided to illustrate how the seal **100** responds to movement of the cable. The cable **302** is shown in three positions relative to the position of the cable at insertion: unmoved (FIG. **3**D), moved in an upward direction (FIG. **3**E), and moved in a downward direction (FIG. **3**F). From these positions, it is possible to imagine the response of the seal **100** to movement of the cable **302** in any direction.

Since the illustration of FIG. **3**C is a side cutaway view showing just a top portion of the cable **302**, a single "slice" of the gradient **326** is indicated. Additionally, the gripper section **226** of the gradient **326** is circled, as this is the portion of the seal that elongates and compresses in response to cable movement, in exemplary embodiments. The seal **100** surrounds the cable **302** and thus the gradient **326** also surrounds the cable **302**. FIGS. **3**D, **3**E, and **3**F show "two slices" of the gradient **326**, indicated as gradient **326***a* and gripper section **226***a* (top of cable **302**) and gradient **326***b* and gripper section **226***b* (bottom of cable).

In FIG. **3**D, the cable **302** is in its insertion position and is unmoved. The ribs **222** of the gradient **326** (F1 position) touch the cable **302** both on top (gradient **326***a*) and on bottom (gradient **326***b*) while neither the gripper section **226***a* (top of cable **302**) nor the gripper section **226***b* (F2 position) touch the cable on either the top or the bottom of the cable. Further, the gripper sections **226***a* and **226***b* do not move, in some embodiments, when the cable is unmoved from its insertion position.

In FIG. **3**E, the cable **302** moves upward, which both pushes against the ribs **222** of gradient **326***a* and creates a space **328***a* between the ribs of gradient **326***b* (F1 position). The space **328***a* creates a risk for ingress of water, dust, and other foreign materials, below the cable **302**, into the fuse box assembly **300**. In exemplary embodiments, the gripper section **226***a* above the cable **302** elongates while the gripper section **226***b* below the cable compresses (F2 position). Recall from FIG. **2**D that there exists spaces **232** between the inside structure **118** and the outside structure **120** of the rubber insert **104**. The compression of the gripper section **226***b* prevents ingress of contaminants underneath the cable **302**, in some embodiments. Further, because the gripper sections **226***a* and **226***b* are opposing sides of the inside structure **118**, the elongation of the gripper section **226***a* facilitates the ability of the gripper section **226***b* to compress, in some embodiments.

FIG. **3**F shows the cable **302** moving in a downward direction, which both pushes against the ribs **222** of gradient **326***b* and creates a space **328***b* between the ribs of gradient **326***a* (F1 position). In exemplary embodiments, the gripper section **226***b* below the cable **302** elongates while the gripper section **226***a* above the cable compresses (F2 position). As two sides of the same cylindrical piece of the rubber insert **104**, the gripper sections **226***a* and **226***b* work together to prevent ingress of foreign matter in response to movement of the cable **302**. Thus, position F1 and position F2 work together to provide an overlap design for sealing the cable **302** inside the fuse box assembly **300**. The design of the rubber insert **104** allows simultaneous elongation on one side and compression on an opposing side of the gripper section **226**, in some embodiments.

In exemplary embodiments, the rubber insert **104** further enables different sizes of cables to be used with the fuse box assembly **300**. The ribs **222** are flexible enough to allow some overlap with the cable. Thus, once inserted, a tight seal is formed between the ribs **222** and the cable. Further, as shown in FIG. **3**C, there is a distance, d, between the inside structure **118** and the cable **302**. It has already been shown in FIG. **2**D that there is enough space **232** for the gripper section **226** of the gradient **326** to move. Larger cables tend to be made of a more rigid material and thus tend to be less flexible than smaller cables. Nevertheless, the overmolded seal **100** is operable with a relatively inflexible and larger cable, in exemplary embodiments, and the IP69 protection is maintained, whatever the cable size.

In exemplary embodiments, the ring **106** of the rubber insert **104** features ring rib **322***a* and ring rib **322***b* (collectively, "ring ribs **322**"). The ring ribs **322** add both surface area and flexibility to the ring **106**, in some embodiments. When the seal **100** is inserted into the opening **324** of the housing **306**, the ring ribs **322** are pushed flush against the housing. Where the housing **306** is made of a harder plastic material and the rubber insert **104** is made of a softer rubber material, the ring ribs **322** facilitate a tight connection between the seal **100** and the housing **306**. Thus, like the ribs **222** and the grippers **228**, the ring ribs **322** are designed to maintain an impenetrable seal to the housing **306** of the fuse box assembly **300** (see also FIG. **3**A). By preventing water ingress into the housing **306** along the gap between the seal **100** and the housing, the position F3 provides IP69 protection that supplements that of position F1 and position F2 for the fuse box assembly **300**, in exemplary embodiments.

FIGS. **4**A-**4**D are representative drawings of a fuse box assembly including a legacy seal design, according to the prior art. FIG. **4**A is an exploded perspective view of a fuse box assembly **400** with a prior art cap assembly **402** and cap seal **404**; FIG. **4**B is a perspective view of the prior art cap seal **404**; FIG. **4**C is a side cutaway view of the cap seal **404**; and FIG. **4**D is a side cutaway view of the prior art fuse box assembly **400**.

The fuse box assembly **400** is able to receive three cables **408**, each including a dedicated terminal **410**, entering from the left side of a housing **406**. The cables **408** are connected to the housing **406** using a cap assembly **402** consisting of three caps, one for each cable, and three separate cap seals **404**. Like the fuse box assembly **300**, the fuse box assembly **400** also includes a cover **412** which is secured to the housing **406** by a cover gasket **414**. The cover gasket **414**, made from a rubber or other elastomeric material, is arranged circumferentially between the housing **406** and the cover **412**, both of which are a harder, plastic material.

The cap seal **404** is designed based on an interference fit in which there is an overlap between the cable **408** and the cap seal. This design is used for small size cables, as the seal is soft without any gaps between the cable **408** and the cap seal **404**. Unfortunately, an ingress of water, dust, and other foreign matter into the fuse box assembly **400** may occur if a larger cable is used because the larger cable is not as flexible as the smaller one. Also, the larger cables tend to be harder than the smaller ones. Also, using the prior art cap seal **404**, an ingress of foreign materials into the fuse box assembly **400** may occur once the cable is off-center. For example, if the cable is bent during assembly or while in use, such as when a vehicle is being driven, the prior art cap seal **404** does not protect the fuse box assembly **400** against ingress of foreign materials.

As illustrated in the perspective view (FIG. **4**B) and the side cutaway view (FIG. **4**C), the cap seal **404** includes three concentric rings **420***a-c* (collectively, "concentric rings **420**") that form an exterior portion **416** of the cap seal, with the interior portion **424** being the receiving chamber for the cable **408**. The interior portion **424** lacks any surface structures, such as the ribs **222** or grippers **228** found in the rubber insert **104** of the novel seal **100**. Further, the structures of the prior art cap seal **404** do not enable elongation of the structures on one side with simultaneous compression on the opposing side seal, as occurs with the novel seal **100** in response to cable movement.

As shown in FIG. **4**A, each cable **408** is fitted with the cap seal **404**, then the cap assembly **402** before being inserted through respective openings of the housing **406**. In the side cutaway view of FIG. **4**B, the cap assembly **402** and cap seal **404** are shown, with the cap seal being partially inside the housing **406** and surrounding the cable **408**. The cap assembly **402** is partially disposed over the housing **406**. The concentric rings **420** press against the housing **406** while the interior portion **418** of the cap seal **404** surrounds the cable **408**. At an end **422** farthest from the terminal **410**, the cap seal **404** is not even touching the cable **408**.

When the cable **408** becomes bent, as indicated by the dashed dotted line, the cap seal **404** is unable to compensate for the cable movement, as occurs with the seal **100**. Instead, dust, dirt, and water can enter the chamber of the fuse assembly **400**, which may affect operation of the fuses inside. Because dust and water, even at high speed and high temperature are present in automotive applications, the prior art cap seal **404** does not provide IP69 protection of the components in the fuse box assembly **400**.

FIG. **5** is a representative drawing of a lock seal assembly **500** used to secure the seal **100** to the fuse box assembly **300**, according to exemplary embodiments. The seal **100** and cable **302** are shown as before, and a portion of the housing **306** is visible. The housing **306** includes an engagement mechanism **502** and a lip **504**, which are either attached or affixed to the housing **306** or formed as a unitary part of the housing, such as by injection molding. Recall from the depiction of the seal **100** (FIGS. **1** and **2**A) that the lock mechanism **110** extends radially outward from the circumferential surface of the plastic body **102** and includes a corner piece **124** that is orthogonal to the lock mechanism piece.

Once the cable is inserted into the housing **306** of the fuse box, the seal **100** will be placed flush against the housing, taking care that the lock mechanism **110** is to the side of the engagement mechanism **502**. In the illustration of FIG. **5**, for example, the lock mechanism **110** would be to the left of the engagement mechanisms **502**. The plastic body **102** is rotated against the housing **306** until the lock mechanism **110** engages with and is latched to the engagement mechanism **502**.

FIGS. **6**A and **6**B are representative drawings of cable assemblies that are supported by the novel seal **100**, according to exemplary embodiments. FIG. **6**A is a perspective view of a cable assembly **600** having a small terminal; FIG. **6**B is a perspective view of a cable assembly **610** having a large terminal. In exemplary embodiments, the novel seal **100** may be used with cables having two different sized terminals.

The cable assembly **600** in FIG. **6**A includes a small terminal **604** connected to cable **602**. The cable assembly **610** in FIG. **6**B includes a larger terminal **614** connected to cable **612**. In some embodiments, cables **602** and **612** are the same size. The larger terminal **614** of is slightly wider at the end distal to the cable **612** whereas the distal end of the small terminal **604** is about the same width as the rest of the terminal. The seal **100** can be used with either the cable assembly **600** or the cable assembly **610**. However, the addition of the seal **100** to each cable is slightly different, as illustrated in FIGS. **7** and **8**.

FIG. **7** includes representative drawings of the process steps for adding the seal **100** (FIG. **1**) to the cable assembly **600** (FIG. **6**A), according to exemplary embodiments. At the left side of FIG. **7**, the seal **100** and the housing of the fuse box assembly **300** (FIG. **3**) are shown. The seal **100** is inserted into an opening of the housing **306**, resulting in a housing with the seal attached **702**. Because the cable assembly **600** has a small terminal, the terminal can fit through the opening of the seal **100**. Thus, the cable assembly **600** with the small terminal is inserted through the seal **100** and into the housing **306**, resulting in a housing **704** with the small terminal attached with the seal.

FIG. **8** includes representative drawings of the process steps for adding the seal **100** (FIG. **1**) to the cable assembly **610** (FIG. **6**B), according to exemplary embodiments. At the left side of FIG. **7**, the seal **100** and the cable **612** (FIG. **6**B) are shown, where the cable is not yet attached to the large terminal **614**. The cable **612** is inserted through the opening of the seal **100**, resulting in a cable with seal attached **802**. The large terminal **614** is then attached to one end of the cable with seal attached **802**, resulting in a cable with seal and large terminal **804**. The cable with seal and large terminal **804** is then fed into an opening of the housing **306**, with the large terminal **614** fitting through the opening. The seal **100** is then secured to the housing **306**, resulting in housing with large terminal attached with seal **806**. Thus, whether the cable uses a smaller terminal or larger one, the assembly of the cable to the fuse box housing is easy for the customer.

The disclosed seal **100** uses an overmolded plastic body with a rubber insert to provide IP69 protection to the fuse box assembly, forming a water-tight seal which prevents the ingress of dust, dirt, and water, even at high pressure and high temperature. The elastic design of the seal **100** maintains the IP69 protection even when the cable is in an off-center position. Further, in exemplary embodiments, the plastic body protects the rubber insert so that it lasts longer and avoids cracking, as the rubber insert is protected from exposure to the foreign matter and particularly protected from high-pressure and/or high-temperature exposure to water. The overmolded seal **100** is easy to use for the customer and facilitates the addition of cables to and the removal of cables from the fuse box assembly. In addition to the fuse box assembly shown and described herein, the overmolded seal **100** can be adapted to other electrical boxes housing electronic equipment, such as power distribution modules.

As used herein, an element or step recited in the singular and proceeded with the word "a" or "an" should be understood as not excluding plural elements or steps, unless such exclusion is explicitly recited. Furthermore, references to "one embodiment" of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features.

While the present disclosure makes reference to certain embodiments, numerous modifications, alterations and changes to the described embodiments are possible without departing from the sphere and scope of the present disclosure, as defined in the appended claim(s). Accordingly, it is intended that the present disclosure not be limited to the described embodiments, but that it has the full scope defined by the language of the following claims, and equivalents thereof.

The invention claimed is:

1. A seal to couple a cable to a fuse box, the seal comprising:

a plastic body comprising a receiving cavity disposed along an inner surface; and

a rubber insert comprising:

an outside structure to join with the inner surface;

an inside structure to surround the cable;

a fitting rib disposed along an outside surface of the outside structure, the fitting rib to occupy the receiving cavity;

a rib disposed circumferentially on an insertion side of the inside structure, the rib to overlap with the cable;

a ring disposed circumferentially around the outside structure and coupled to the fitting rib;

one or more ring ribs extending from the ring toward the fuse box, wherein the one or more ring ribs fit tightly against a surface of the fuse box; and

a gripper section disposed circumferentially on a terminal side of the inside structure, the gripper section to overlap with the cable, wherein the gripper section comprises first, second, and third grippers, the rib, the first gripper, the second gripper, and the third gripper to form a gradient that provides IP**69** protection of the fuse box;

wherein the inside structure of the rubber insert moves in response to movement of the cable while the outside structure does not move.

2. The seal of claim **1**, wherein the plastic body and the rubber insert are overmolded to form a unitary structure.

3. The seal of claim **1**, the plastic body further comprising a lock mechanism disposed along a circumferential edge.

4. The seal of claim **1**, further comprising:

second, third, and fourth receiving cavities; and

second, third, and fourth fitting ribs;

wherein the second fitting rib occupies the second receiving cavity, the third fitting rib occupies the third receiving cavity, and the fourth fitting rib occupies the fourth receiving cavity.

5. The seal of claim **1**, wherein the gripper section simultaneously elongates on one side of the cable and compresses on an opposing side of the cable in response to movement of the cable while the outside structure does not move.

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